

/ Transfer Learning:

A. the definition of Transfer Learning:

it involves taking a pretrained model or parts of a pretrained model that are trained for one task, and using them to solve another different, but related task.
the idea is that some of the skills learned for the original task transfer and are applicable to the new task.

B. the types of Transfer Learning:

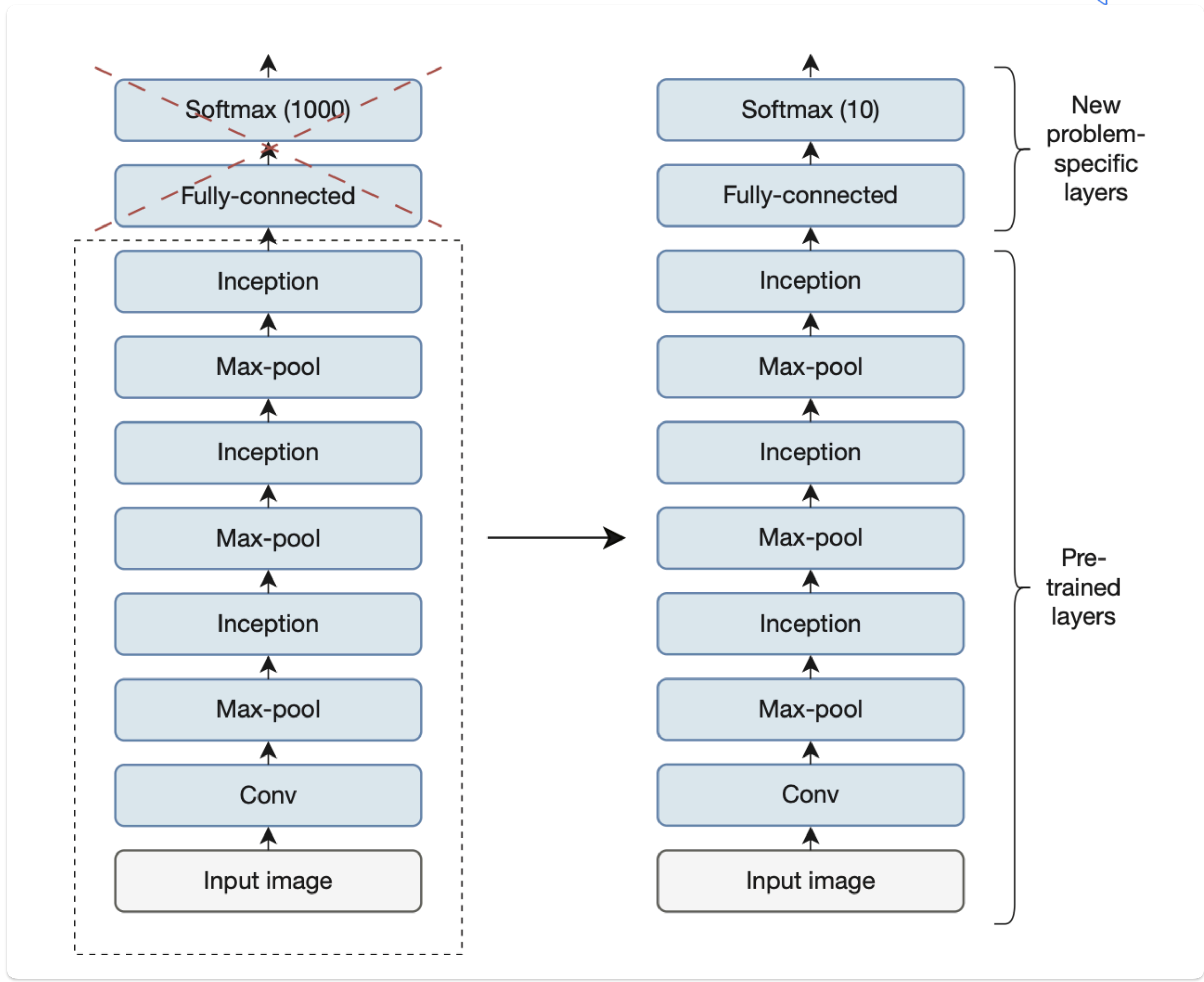
- ⇒ Fine Tuning.
- ⇒ Feature Extraction.

2. Transfer Learning — Fine Tuning:

- A. it starts with a pretrained model for one task, and then training this pretrained model with our own data for another different, but related task.
- B. this is often beneficial if our own dataset is limited size.
even if we have a large dataset, starting from a pretrained model can still be beneficial because it can reduce the amount of time you need to spend train with your own dataset.

3. Transfer Learning — Feature Extraction:

- A. it is to use parts of a pretrained model that are trained for one task as a building block in our own model intended to solve another different, but related task.
- B. in this example, we could use a pretrained model and replace some of the last layers with our own layers end with a ten-output softmax layer.
furthermore, the surplus pretrained convolutional layers already have the ability to recognize specific features that are useful for identifying different types dogs.
we would then train this model on our own data with the ten types of dogs we want this model to classify.



C. the freezing weights session:

- ⇒ if we just go ahead and start training on our own dataset in our model ,
there is a risk that the learning algorithm ruins the carefully pretrained weights in the pretrained part of our model.
therefore , it is often a good idea to freeze these pretrained weights and only train the new weights in our newly added layers in the early training stage.
- ⇒ this also makes the training process faster because the number of adjustable weights is significantly smaller.

D. the unfreezing weights session:

after training our model for the early several epochs with the pretrained weights frozen ,
the next step can be to keep training our model with the unfrozen pretrained weights for another few epochs with a smaller learning rate.